# Aggregating Markups in Production Networks

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**Abstract**

How should firm-level markups be aggregated into an economy-wide measure of market power when production takes place through input–output networks? We show that commonly used aggregation methods do not recover the economic incidence of market power, either because they focus on final sales or because they rely on gross transaction weights that do not reflect cost-based exposure. We derive the aggregate markup consistent with household incidence, defined as the ratio of total household expenditure to the primary-factor cost of producing the observed consumption bundle. This aggregate markup admits an exact geometric representation with cost-based Domar weights that trace how distortions propagate through the production network. Using near-universe Belgian firm-to-firm data for 2002–2014, we find that the aggregate markup is modest and stable, ranging from 1.16 to 1.19 with a slight upward drift, and that upstream firms account for about one quarter to one third of the total price–cost wedge. These results show that market power is shaped by production networks, and that aggregators that ignore them mismeasure both the level and the sources of aggregate distortions.

**Keywords:** Markups, Aggregation, Production Networks, Multiple Marginalization, Input-Output, Domar Weights
**JEL:** D22, D24, D43, L11, L13, O47

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# 1 Introduction

Markups drive a wedge between prices and marginal costs, shaping resource allocation, household prices, and welfare. A large empirical literature has documented substantial heterogeneity in firm-level markups and linked that heterogeneity to productivity, competition, and market power. Yet a basic question remains unresolved: how should these micro-level markups be aggregated into an economy-wide measure of market power?

This question is nontrivial because production is organized as a network. Goods and services typically pass through multiple firms before reaching final demand, so markups applied upstream raise the cost of intermediate inputs and are then marked up again downstream. In such an environment, aggregate markups depend not only on the level of firm-level markups, but also on where firms sit in the production network and how their distortions propagate through supply chains.

Commonly used aggregation methods are not designed to recover the cumulative price–cost wedge implied by this network structure. Final-demand-based measures capture only pricing at the point of sale and therefore omit upstream distortions embodied in intermediate inputs. Revenue-based measures go in the opposite direction: because gross revenues record the repeated passage of goods through production chains, they overweight firms whose output appears multiple times in downstream transactions. As a result, existing measures do not recover the economic incidence of market power in network economies.

This paper develops a micro-founded framework that identifies the exact aggregate markup consistent with household incidence in production networks. We define the aggregate markup as the ratio of total household expenditure to the primary-factor cost required to produce the observed consumption bundle. Denoting this object by $\mu^\star$, we show that it is the unique scalar aggregator of firm-level markups consistent with that incidence concept. In this sense, $\mu^\star$ is not merely a summary statistic for the cross-section of firm-level markups. It is the exact cumulative price–cost wedge faced by households once upstream propagation through the production network is taken into account.

Our main theoretical result is that $\mu^\star$ admits an exact multiplicative representation. Specifically, the aggregate markup can be written as a geometric average of firm-level markups, with weights given by cost-based Domar weights. These weights measure how much of the primary-factor cost of final demand flows through each firm, accounting for both direct and indirect input linkages. Unlike revenue-based weights, they track economic incidence rather than gross transaction values. The framework also yields a decomposition of the aggregate markup by production distance, which makes it possible to distinguish distortions applied at the point of final sale from those originating upstream and transmitted through intermediate input linkages.

We implement the framework using near-universe Belgian administrative data covering the full network of domestic firm-to-firm transactions over the period 2002–2014. Three main empirical

results emerge.

First, the network-consistent aggregate markup is modest and remarkably stable over time. The implied economy-wide markup fluctuates in a narrow range between 1.16 and 1.19, with only a slight upward drift over the sample period.

Second, the choice of aggregation method matters quantitatively. Standard sales-, cost-, and value-added-weighted averages differ systematically from $\mu^\star$, the geometric final-demand measure is lower because it excludes upstream distortions, and the revenue-based Domar aggregator substantially overstates aggregate markups because it places excessive weight on gross marked-up flows. The network-consistent measure $\mu^\star$ lies between these alternatives, exactly as predicted by the theory.

Third, upstream production plays a quantitatively meaningful role in aggregate market power. While most of the aggregate markup originates at the final stage of production, upstream production layers account for roughly one quarter to one third of the total price–cost wedge. This highlights the role of network propagation in shaping the aggregate price–cost wedge faced by households.

These findings imply that market power is inherently a network phenomenon. The aggregate impact of a firm's markup depends not only on its level, but also on its position in the production network and on the extent to which its output is used in downstream production. Measures based solely on final sales or gross revenues therefore risk mischaracterizing both the magnitude and the origin of aggregate distortions.

The paper makes two contributions. First, it provides an exact aggregation framework for firm-level markups in production networks and identifies the unique aggregate object consistent with household incidence. This clarifies that commonly used aggregators do not measure the same economic object and therefore cannot be interpreted as interchangeable measures of aggregate market power. Second, it shows empirically that accounting for network structure matters quantitatively for the measurement of aggregate market power, while also showing that the relevant propagation is concentrated in the first few upstream layers. This combination of theoretical discipline and empirical tractability makes the framework useful both for measurement and for future work on the welfare consequences of market power in production networks.

The intuition for the main result is straightforward. Along any given production chain, markups compound multiplicatively: if an upstream firm marks up an input and a downstream firm marks up the resulting cost, the total distortion is the product of the two markups. Aggregate markups therefore reflect products of firm-level markups across production paths, with cost-based Domar weights determining how strongly each firm's markup affects final demand through the production network.

The remainder of the paper is organized as follows. Section 2 presents the aggregation framework. Section 3 studies the properties of the aggregate markup and relates it to standard measures. Section 4 describes the data. Section 5 reports the empirical results. Section 6 concludes.

## 1.1 Related Literature

This paper studies how firm-level markups map into an aggregate measure of market power in an economy with production networks. While a large literature has measured markups at the firm level, studied their welfare implications, and analyzed how distortions propagate through input–output linkages, these strands have largely evolved in parallel. As a result, a basic question remains unresolved: what is the exact aggregate object that firm-level markups imply once production is organized as a network? This paper brings these literatures together and shows that the answer is a unique, network-consistent aggregator that captures the cumulative price–cost wedge faced by households.

First, we build on the production-approach literature that estimates firm-level markups from output elasticities and expenditure shares, following De Loecker and Warzynski (2012). Using this methodology, De Loecker, Eeckhout, and Unger (2020) document a rise in aggregate markups in the United States, while De Loecker and Eeckhout (2018) provide global evidence across countries and firms. This literature has greatly improved the measurement of *firm-level* market power, but it leaves open how these markups should be mapped into an *aggregate* object when firms are linked through intermediate-input trade. Standard weighted averages summarize the cross-section of firm markups, but they are not designed to measure the cumulative price–cost wedge borne by households in a production network. Our paper addresses this aggregation problem by deriving the exact mapping from firm-level markups to the household price–cost wedge implied by the production network.

Second, our paper relates to work studying aggregate markups and their welfare implications. Edmond, Midrigan, and D. Y. Xu (2015) emphasize the role of markup dispersion for aggregate distortions, while Edmond, Midrigan, and Y. Xu (2023) quantify the welfare costs of markups in general equilibrium. These papers clarify why the aggregation of market power matters, but they do not derive a network-consistent aggregator for economies in which markups propagate through firm-to-firm production linkages. We complement this literature by showing that once production is organized as a network, the relevant aggregate object is multiplicative and must be weighted by cost-based rather than revenue-based exposure to final demand.

Third, our paper relates to the literature on vertical market power and multiple marginalization, starting with Spengler (1950). A large body of work studies how markups compound along supply chains and how contractual frictions shape the distribution of rents across vertically related firms (Mathewson and Winter 1984; Bonanno and Vickers 1988; Rey and Tirole 2007; Durán, Micco, and Perloff 2022). Empirical studies document these mechanisms in particular industries, including retail, video rental, food, and television markets (Villas-Boas 2007; Mortimer 2008; Bonnet and Dubois 2010; Crawford et al. 2018). We extend this logic from specific vertical chains to the entire production network. In our setting, the question is not only whether double marginalization arises within a given chain, but how all such wedges aggregate into a single economy-wide object once firms are embedded in a general production network.

Fourth, our paper is closely related to recent attempts to map micro markups into consumer or network-based aggregates. Grassi (2017) studies markup aggregation in an input–output environment, while Ganglmair, Kann, and Tsanko (2021) approximate households' exposure to markups using final-demand weights. These papers are particularly close in spirit to ours, but they stop short of deriving the exact network-consistent aggregator that corresponds to the cumulative household price–cost wedge. Our framework clarifies why this object differs from both final-demand averages and revenue-based network aggregates: the former omit upstream propagation, while the latter track gross marked-up flows rather than the underlying primary-factor costs relevant for economic incidence.

Fifth, our paper builds on the production-network literature. Hulten (1978) shows that aggregate productivity growth can be expressed as a Domar-weighted sum of firm-level changes, a result later interpreted as a welfare-accounting identity by Petrin and Levinsohn (2012). Baqaee and Farhi (2019) and Baqaee and Farhi (2020) extend this logic to general equilibrium environments with distortions and nonlinear propagation, while related work studies how input–output linkages shape aggregate fluctuations and the transmission of shocks (Acemoglu et al. 2012; Carvalho et al. 2021). We apply this network-aggregation logic to markups rather than productivity. The key distinction is that, unlike productivity, markups distort prices multiplicatively along production chains. As a result, the relevant aggregate object is not a revenue-based Domar average, but a cost-based geometric aggregator that captures how distortions compound and propagate to final demand. This distinction is central for measurement because it separates gross-output exposure from welfare-relevant incidence.

Finally, our results speak to the literature on global value chains and value capture across production stages. This literature emphasizes that positions along the value chain are associated with systematic differences in rents, value added, and control, with downstream activities often capturing a disproportionate share of value because of differentiation, branding, and proximity to final demand (Antràs and Chor 2013; Antràs 2020; Baldwin and Ito 2021). Our paper is complementary to this perspective. Rather than studying the distribution of value added, we study the distribution and aggregation of markup wedges. The empirical finding that most of the aggregate markup originates at or near the final stage of production, while upstream layers remain quantitatively meaningful, provides a network-based counterpart to this broader view of value capture along production chains.

Taken together, our contribution is to provide a unified framework that connects firm-level markup measurement to network-based aggregation and household incidence. While existing approaches capture either the distribution of markups, their welfare implications, or their propagation through networks, our framework identifies the exact aggregate object implied by all three. This clarifies that commonly used aggregators correspond to different economic concepts and therefore cannot be interpreted as interchangeable measures of aggregate market power.

# 2 Aggregation Framework

The object of interest is the aggregate markup implied by household incidence in a production network. The question is not how to summarize the cross-section of firm-level markups, but which scalar object captures the cumulative price–cost wedge borne by households once markups propagate through intermediate input linkages. In a network economy, this distinction is essential: because goods pass through multiple production stages before reaching final demand, upstream markups affect household prices even when they are not applied at the point of final sale.

We therefore seek an aggregate object that maps firm-level markups $\{\mu_i\}_i$ into a scalar $\mu^\star$ with a clear economic interpretation. Specifically, $\mu^\star$ should measure the ratio of total household expenditure to the primary-factor cost required to produce the observed final-demand bundle. This requirement pins down the relevant notion of economic incidence and will imply a unique network-consistent aggregation of firm-level markups.

## 2.1 Environment and Objects

Consider an economy with $N$ firms linked through a production network. Firms produce differentiated goods that can be used either as intermediate inputs or sold to final consumers. Production may involve multiple stages, so that goods pass through several firms before reaching final demand.

**Assumption 1** (Baseline)**.** *Technologies exhibit constant returns to scale. Firms set markups that are invariant to quantities, prices, and trading partners, but may vary across firms and over time. Primary-factor markets are perfectly competitive, and preferences over final goods are homothetic.*

**Firm markups.** Each firm $i$ charges a constant markup $\mu_i$ over marginal cost. Following De Loecker and Warzynski (2012), markups are recovered from production data as

$$\mu_i = \frac{\varepsilon_{X_i,Y_i}}{a_{X_i,Y_i}},$$

where $\varepsilon_{X_i,Y_i}$ is the output elasticity with respect to a flexible input and $a_{X_i,Y_i}$ is the corresponding expenditure share. Markups are firm-specific but do not vary across transactions.

**Production network.** Let $\Omega$ denote the revenue-based input–output matrix, where

$$\Omega_{ij} = X_{ij}/R_i$$

is the share of firm $i$'s revenue spent on inputs from supplier $j$. The matrix $\Omega$ summarizes the direct input requirements of production.

To incorporate indirect linkages, define the associated Leontief inverse

$$\Psi \equiv (I - \Omega)^{-1}.$$

The matrix $\Psi$ captures the total input requirements embodied in one unit of revenue, including both direct suppliers and the suppliers of those suppliers throughout the network.

**Final demand.** Let $\mathbf{b}$ denote the vector of final-demand expenditure shares, where $b_i$ is the share of total final expenditure spent on firm $i$'s output. This vector identifies where production chains terminate and therefore provides the natural starting point for measuring household incidence.

**Why revenue-based exposure is not enough.** Combining $\mathbf{b}$ with the Leontief inverse yields the standard Domar weights,

$$\lambda' \equiv \mathbf{b}'\Psi.$$

These weights measure how much gross production associated with final demand flows through each firm, directly or indirectly. They are the natural object for productivity analysis because productivity shocks propagate through gross output.

For markup aggregation, however, gross-output exposure is not the relevant concept. The economic question is not how much gross revenue passes through a firm, but how much of the underlying primary-factor cost of final demand is exposed to that firm's markup. When markups are present, gross revenues already embody upstream distortions and therefore reflect marked-up transaction values rather than the underlying cost base relevant for household incidence. Revenue-based weights are therefore not incorrect as accounting objects; they simply do not recover the incidence concept that an aggregate markup is meant to capture.

**Cost-based input–output structure.** To recover the relevant incidence concept, we construct a cost-based representation of the production network. Define the markup-adjusted input–output matrix

$$\widetilde{\Omega} \equiv \operatorname{diag}(\mu)\,\Omega,$$

and the associated Leontief inverse

$$\widetilde{\Psi} \equiv (I - \widetilde{\Omega})^{-1}.$$

We then define cost-based Domar weights as

$$\widetilde{\lambda}' \equiv \mathbf{b}'\widetilde{\Psi}.$$

These weights measure how much of the primary-factor cost required to produce final demand flows through each firm. Relative to revenue-based weights, they replace gross transaction exposure

with cost exposure and therefore capture how strongly each firm's markup contributes to the final household price bundle. This is the relevant weighting object for aggregate markups.

## 2.2 Prices, Costs, and the Aggregate Markup

We now characterize the aggregate object implied by household incidence in the network economy.

**Prices and cost propagation.** Let $\mathbf{w}$ denote the vector of primary-factor prices, which we take as the numeraire. Under Assumption 1, equilibrium prices reflect both direct production costs and the cumulative effect of upstream markups embodied in intermediate inputs. Using the cost-based Leontief inverse, equilibrium producer prices can be written as

$$p = \operatorname{diag}(\mu)\, \widetilde{\Psi}\mathbf{w}.$$

This expression makes clear that final prices incorporate the full sequence of upstream distortions: $\widetilde{\Psi}$ propagates primary-factor costs through the network, while $\operatorname{diag}(\mu)$ applies firm-level markups at each stage of production.

**Aggregate expenditure and cost.** Define

$$E \equiv \mathbf{b}'p \qquad \text{and} \qquad F \equiv \mathbf{b}'\widetilde{\Psi}\mathbf{w},$$

where $E$ is total household expenditure on the observed final-demand bundle and $F$ is the total primary-factor cost required to produce that same bundle. The distinction is central. Household expenditure reflects all markups applied along production chains, whereas $F$ isolates the underlying factor cost net of those distortions.

**Definition 1** (Aggregate markup)**.** The aggregate markup is defined as

$$\mu^\star \equiv \frac{E}{F}.$$

**Interpretation.** The object $\mu^\star$ is the aggregate markup consistent with household incidence. It measures the cumulative wedge between what households pay for the observed consumption bundle and the primary-factor cost required to produce it. Unlike standard averages of firm-level markups, $\mu^\star$ is not merely a descriptive statistic for the markup distribution. It is the exact economy-wide price–cost wedge implied by the observed vector of firm-level markups and the observed production network.

This definition also makes clear why the aggregate markup is unique. Once the target object is specified as the ratio of household expenditure to the primary-factor cost of producing the same bundle, there is only one scalar aggregator that recovers it. Alternative aggregation schemes may

summarize firm-level markups in useful ways, but they correspond to different economic objects because they either omit upstream propagation or weight firms by exposure concepts that do not match household incidence.

### 2.3 Domar Weights and Multiplicative Aggregation

Having defined the aggregate markup $\mu^\star$ as the ratio of household expenditure to primary-factor cost, we now characterize how firm-level markups map into this object. The central question is immediate: once markups propagate through a production network and compound along production chains, what is the exact aggregation of $\{\mu_i\}_i$ that recovers $\mu^\star$?

The answer reflects two fundamental features of production networks. First, markups compound multiplicatively along production chains. Second, firms differ in how strongly their markups affect final demand, depending on their position in the network and the extent to which their output is used downstream. These two forces jointly determine both the functional form of aggregation and the appropriate weights.

**Multiplicative propagation.** To see why aggregation is multiplicative, consider a simple two-stage production chain. An upstream firm produces an intermediate input with markup $\mu_1$, which is then used by a downstream firm that applies markup $\mu_2$ before selling to consumers. The upstream markup raises the downstream firm's marginal cost, and the downstream firm marks up that higher cost. The total distortion is therefore $\mu_1 \times \mu_2$, not $\mu_1 + \mu_2$.

This logic extends directly to general production networks. Along any production path linking a firm to final demand, the total price–cost wedge is the product of the markups encountered along that path. Aggregating across all such paths therefore requires averaging products of markups rather than summing them. This implies that the aggregate markup must be a geometric, or equivalently log-linear, aggregation of firm-level markups.

**Cost-based exposure.** Once aggregation is multiplicative, the remaining question is how much weight to assign to each firm. The relevant weights must capture how strongly a firm's markup affects the primary-factor cost of final demand. This depends not only on whether the firm sells to households, but also on how intensively its output is used in downstream production and how far its distortions propagate through the network.

The cost-based Domar weights $\widetilde{\lambda}_i$ provide exactly this object. By construction, $\widetilde{\lambda}_i$ measures the share of the primary-factor cost required to produce final demand that flows through firm $i$, accounting for both direct and indirect input linkages. A firm that supplies inputs to many downstream producers, or that is located upstream in long production chains, receives a larger weight because its markup is embedded in many downstream prices.

**Proposition 1** (Exact aggregation of firm-level markups)**.** *Under Assumption 1, the aggregate markup $\mu^\star$ admits the exact representation*

$$\ln \mu^\star = \sum_i \widetilde{\lambda}_i \ln \mu_i, \qquad \widetilde{\boldsymbol{\lambda}}' = \mathbf{b}'(I - \widetilde{\Omega})^{-1}, \qquad \widetilde{\Omega} = \operatorname{diag}(\mu)\,\Omega.$$

*Equivalently,*

$$\mu^\star = \prod_i \mu_i^{\widetilde{\lambda}_i}.$$

*The weights $\widetilde{\lambda}_i$ are cost-based Domar weights and measure each firm's total exposure to the primary-factor cost of final demand, accounting for both direct and indirect production linkages.*

*Moreover, local comparative statics satisfy*

$$d \ln \mu^\star = \sum_i \widetilde{\lambda}_i \, d \ln \mu_i.$$

*Proof.* See Appendix A.3. □

**Interpretation.** Proposition 1 shows that the aggregate markup is a geometric aggregation of firm-level markups, with weights given by cost-based Domar weights. Each firm's contribution reflects how much of the primary-factor cost of final demand is exposed to its markup once all production paths are taken into account.

Two implications follow. First, upstream firms contribute to the aggregate markup even if they do not sell directly to households, because their markups are embodied in intermediate inputs used downstream. Second, each firm's markup enters the aggregate exactly once in incidence terms, despite appearing multiple times in gross transactions.

**Why standard aggregators differ.** Proposition 1 clarifies why commonly used aggregation schemes do not recover the aggregate markup. Arithmetic averages impose an additive structure that is inconsistent with multiplicative compounding, while revenue-based weights track gross marked-up flows rather than the underlying cost exposure relevant for household prices. As a result, these measures correspond to different economic objects. By contrast, $\mu^\star$ is uniquely pinned down by the requirement that it equates household expenditure to the primary-factor cost of producing the observed consumption bundle.

## 3 Properties of the Aggregate Markup

This section studies the economic content of the aggregate markup and clarifies how it differs from existing measures. We first show that the aggregation result admits a decomposition across production distance, which provides a structural notion of incidence in the production network. We

then compare the proposed aggregator to commonly used alternatives in the literature and show that they fail to measure the cumulative price–cost wedge faced by households.

### 3.1 Decomposition by Production Distance

While Proposition 1 characterizes how firm-level markups aggregate, it does not reveal how markup distortions propagate through the production network. In particular, it remains unclear whether aggregate distortions are primarily driven by firms close to final demand or by upstream suppliers.

The following result decomposes the aggregate markup into contributions associated with different distances to final demand.

**Proposition 2** (Path decomposition)**.** *The aggregate markup satisfies*

$$\ln \mu^\star = \sum_{L=0}^{\infty} \mathbf{b}' \widetilde{\Omega}(\mu)^L \ln \mu,$$

*where* $\widetilde{\Omega}(\mu) = \operatorname{diag}(\mu)\,\Omega$.

*Define the normalized contribution of layer* $L$ *as*

$$\Upsilon_L \equiv \frac{\mathbf{b}' \widetilde{\Omega}(\mu)^L \ln \mu}{\ln \mu^\star}.$$

*Then*

$$\sum_{L \geq 0} \Upsilon_L = 1.$$

*Proof.* See Appendix A.4. □

**Interpretation.** The decomposition expresses the aggregate markup as a sum of contributions from successive upstream layers of production. The term with $L = 0$ captures markups of firms selling directly to final demand, while higher-order terms reflect markups embodied in intermediate inputs supplied by increasingly upstream firms.

This representation provides a structural notion of incidence across production distance. Rather than focusing on which firms have high markups, it identifies how distortions are transmitted to households through the production network. Because the weights decay geometrically, the decomposition also shows that aggregate markup distortions are primarily driven by firms located close to final demand.

### 3.2 Comparison with Benchmark Aggregators

We now relate the proposed aggregator $\mu^\star$ to commonly used measures of aggregate markups in the literature. Existing approaches differ in whether they account for production networks and

whether they respect the multiplicative nature of markup compounding. We present six benchmark aggregators that correspond to different notions of exposure.

**Benchmark aggregators. (a) Sales-weighted arithmetic aggregate**

$$\mu^R \equiv \sum_i \omega_i^R \mu_i, \qquad \omega_i^R \equiv \frac{R_i}{\sum_j R_j}.$$

Widely used in the empirical markup literature (e.g. De Loecker, Eeckhout, and Unger 2020), this measure reflects gross revenue shares. Because revenues include repeated revaluation of intermediate goods, it overweights firms in long production chains relative to their contribution to final demand. In addition, revenue weights are themselves affected by markups: firms with higher markups mechanically generate higher revenues, which increases their weight in the aggregation. As a result, the weighting scheme is endogenous to the object being aggregated and amplifies the contribution of high-markup firms.

**(b) Cost-weighted arithmetic aggregate**

$$\mu^C \equiv \sum_i \omega_i^C \mu_i, \qquad \omega_i^C \equiv \frac{C_i}{\sum_j C_j}.$$

Used as an alternative benchmark (e.g. Edmond, Midrigan, and Y. Xu 2023), cost weights reduce the amplification present in revenue weights. However, costs themselves embed markups applied by upstream firms through intermediate input prices. As a result, cost weights incorporate markup propagation only indirectly and in a way that is not aligned with economic incidence. They therefore mix different layers of the production network without properly accounting for how markups accumulate along production chains.

**(c) Value-added-weighted aggregate**

$$\mu^{VA} \equiv \frac{\sum_i VA_i \mu_i}{\sum_i VA_i}.$$

Value-added weights net out intermediate inputs but do not map to final demand exposure and therefore miss upstream propagation effects.

**(d) Final-demand arithmetic aggregate**

$$\mu^{FD,A} \equiv \sum_{i \in FD} b_i \mu_i.$$

This measure focuses on firms selling directly to final consumers. It is closely related to the object approximated by Ganglmair, Kann, and Tsanko (2021). However, it omits upstream firms whose markups are embodied in intermediate inputs.

**(e) Final-demand geometric aggregate**

$$\mu^{FD,G} \equiv \prod_{i \in FD} \mu_i^{b_i}.$$

This is the geometric counterpart of $\mu^{FD,A}$ and provides the natural benchmark for comparison with $\mu^\star$, as both respect multiplicative aggregation.

Using the Neumann expansion of the cost-based Leontief inverse,

$$\widetilde{\Psi} = I + \widetilde{\Omega} + \widetilde{\Omega}^2 + \cdots,$$

we can write

$$\ln \mu^\star = \underbrace{\sum_{i \in FD} b_i \ln \mu_i}_{\ln \mu^{FD,G}} + \sum_{L=1}^{\infty} b' \widetilde{\Omega}^L \ln \mu.$$

Because all terms are nonnegative, it follows that

$$\mu^{FD,G} \leq \mu^\star.$$

*Interpretation.* The geometric final-demand aggregator corresponds to the direct markup applied at the point of final sale. The network aggregator $\mu^\star$ augments this with upstream contributions transmitted through production chains. The gap between the two measures therefore captures the cumulative effect of upstream markups embodied in final goods.

**(f) Revenue-based Domar (geometric) aggregate**

$$\mu^D \equiv \prod_i \mu_i^{\lambda_i}, \qquad \lambda' \equiv \mathbf{b}'\Psi.$$

This aggregator builds on revenue-based Domar weights (e.g. Hulten 1978; Petrin and Levinsohn 2012) and is appropriate for productivity analysis. However, because it is based on gross output flows, it overstates the importance of upstream firms for markup aggregation.

**Economic differences.** The benchmark aggregators can be understood as measuring different notions of exposure. Measures based on accounting flows ($\mu^R$, $\mu^C$, $\mu^{VA}$) weight firms using observed transactions, but these flows reflect marked-up values and therefore do not correspond to economic incidence. Final-demand measures ($\mu^{FD,A}$, $\mu^{FD,G}$) instead isolate firms selling directly to consumers, but abstract from the propagation of markups through intermediate input linkages. Revenue-based Domar aggregation ($\mu^D$) incorporates network structure, but relies on gross-output weights that track marked-up flows rather than primary-factor cost exposure.

By contrast, $\mu^\star$ measures the aggregate markup as the price–cost wedge faced by households.

Its cost-based Domar weights

$$\widetilde{\lambda}' = \mathbf{b}'\widetilde{\Psi}(\mu)$$

map firm-level markups into this object by tracing how primary-factor costs propagate through the production network. This ensures that markups are aggregated consistently with economic incidence, rather than with accounting flows or partial network exposure.

# 4 Data

Our empirical analysis uses the Belgian microdata infrastructure developed by the National Bank of Belgium (NBB), following Dhyne, Ayumu K. Kikkawa, and Magerman (2022). The data combine three complementary administrative sources: (i) firms' annual accounts, (ii) international trade records, and (iii) the universe of domestic business-to-business (B2B) transactions reported for value-added tax (VAT) purposes.

A key feature of this setting is that VAT declarations provide a near-universe mapping of domestic inter-firm transactions. This allows us to reconstruct the production network at the firm level and to trace how distortions propagate through supply chains. Such comprehensive firm-to-firm data are rare and essential for implementing a network-consistent aggregation of markups. This structure allows us to map the empirical objects directly to the theoretical framework in Section 2: firm-to-firm transactions identify the input–output matrix $\Omega$, firm accounts recover markups $\mu_i$, and final-demand shares $b$ are constructed from sales to final users.

This dataset has been widely used to study firm-to-firm trade and production networks. Dhyne, Ayumu K. Kikkawa, and Magerman (2022) document imperfect competition in firm-to-firm trade, while Dhyne, Ayumu Ken Kikkawa, Mogstad, et al. (2021) and Bernard et al. (2022) analyze the structure of production networks. Recent work uses the same data to study trade exposure and demand shocks (Dhyne, Ayumu Ken Kikkawa, Mogstad, et al. (2023) and Dhyne, Ayumu Ken Kikkawa, Komatsu, et al. (2025)).

**VAT B2B registry.** The backbone of the dataset is the VAT registry of inter-firm invoices, which records essentially all domestic B2B transactions exceeding €250 per year.[1] Each record identifies the buyer, the supplier, and the transaction value. Aggregating these flows at the annual frequency yields a near-complete buyer–seller network covering both goods and services. We use these transactions to construct the revenue-based input–output matrix $\Omega$, where each entry measures the share of a firm's expenditure allocated to each supplier.

**Firm accounts and trade data.** We merge the VAT registry with firms' annual accounts using NBB enterprise identifiers. The accounts provide detailed information on employment, payroll,

[1] As documented in Dhyne, Magerman, and Rubínova (2015), the €250 threshold excludes only a small fraction of micro-invoices and has negligible impact on aggregate linkage patterns.

tangible assets, intermediate inputs, and turnover, which we use to estimate firm-level markups and measure primary-factor usage. These variables allow us to estimate firm-level markups following De Loecker and Warzynski (2012), consistent with the definition of $\mu_i$ in Section 2.

We further merge international trade data from customs declarations, which report firm-level imports and exports. Imports are treated as purchases from an external sector with markup normalized to one, so that foreign suppliers do not introduce additional distortions within the domestic production network, while exports allow us to distinguish domestic from foreign sales when constructing the input–output structure.

The merged dataset yields, for each firm-year, a comprehensive profile of domestic purchases, domestic sales, imported inputs, exported outputs, and primary-factor usage. Coverage is extensive, accounting for more than 95% of Belgian value added and about 90% of domestic inter-firm transactions.

**Sample and restrictions.** We focus on the period 2002–2014, during which data coverage and reporting standards are stable. We restrict attention to private, non-financial firms with at least one full-time equivalent employee and at least €100 of tangible assets, following Dhyne, Magerman, and Rubínova (2015).

To mitigate the influence of outliers, we trim the top and bottom one percent of the markup distribution, following De Loecker and Warzynski (2012) and De Loecker, Eeckhout, and Unger (2020).

**Summary.** The resulting panel combines near-universe coverage of domestic production linkages with detailed firm-level information and complete trade data. This allows us to construct the empirical counterparts of the theoretical objects in Section 2, including the input–output matrix $\Omega$, firm-level markups $\mu_i$, and final-demand shares $b$. Using these objects, we compute cost-based input–output matrices and Domar weights, which capture how markups propagate through the production network.

This level of granularity is crucial for measuring aggregate markups in a manner consistent with the structure of production networks.

# 5 Empirical Results

This section implements the theoretical aggregation framework using firm-level data and input–output linkages. Guided by the results in Section 3, the empirical analysis focuses on three objects. First, we measure the aggregate markup $\mu^\star$ and compare it to commonly used benchmark aggregators. Second, we quantify the gap between $\mu^\star$ and the geometric final-demand aggregator $\mu^{FD,G}$, which isolates the role of upstream propagation. Third, we decompose $\mu^\star$ by production distance

using the path decomposition in Proposition 2, which provides a structural measure of incidence across the production network.

### 5.1 Firm-Level Markups: Levels and Distribution

We begin by describing the firm-level markups that enter the aggregation. These objects constitute the primitive inputs into the network-based measure $\mu^\star$, and their distribution plays a central role in determining how distortions propagate through the production network.

We estimate markups $\mu_i$ using the production approach of De Loecker and Warzynski (2012). Specifically, we treat Cost of Goods Sold (COGS) as the flexible input and set the output elasticity equal to the median cost share at the sector-year level, following De Loecker, Eeckhout, and Unger (2020). A markup $\mu_i = 1.1$ corresponds to a 10% wedge between price and marginal cost.

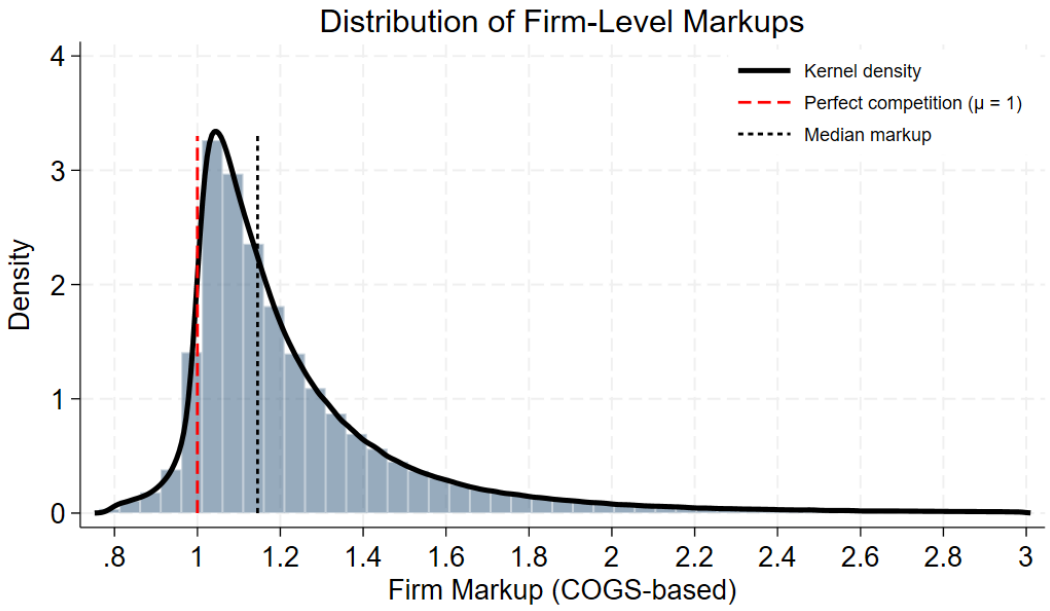



Figure 1: Distribution of firm-level markups

Figure 1 shows that the distribution of firm-level markups is right-skewed and concentrated slightly above one. Most firms exhibit modest markups in the range of roughly 1.0 to 1.3, while a smaller set of firms in the right tail display substantially higher markups.

Table 1: Average Markup

| Arithmetic | Sales-weighted | Sales-to-HH-weighted | Cost-weighted | VA-weighted |
|---|---|---|---|---|
| 1.25 | 1.19 | 1.14 | 1.14 | 1.21 |

Table 1 reports standard aggregation schemes over the 2002–2014 period. The unweighted

mean markup is 1.25, while weighted averages are systematically lower, ranging from 1.14 to 1.21 depending on the weighting scheme. This reflects the fact that high-markup firms tend to receive relatively small weights under most aggregation methods.

These patterns highlight two features that are central for aggregation. First, heterogeneity in markups is substantial, implying that the mapping from firm-level markups to aggregate objects is quantitatively non-trivial. Second, the choice of weights matters even in the absence of network considerations, as different weighting schemes place different emphasis on high-markup firms.

Importantly, firm-level markups do not map one-for-one into the markup faced by households. Two firms with identical markups can have very different contributions to the aggregate markup depending on their position in the production network and on how intensively their output is used in downstream production. The next subsections show how the network structure governs this mapping and determines the aggregate price–cost wedge.

### 5.2 The Aggregate Markup Over Time

We now construct the economy-wide markup using the network-consistent aggregator derived in Section 3:

$$\mu_t^\star = \prod_i \mu_{it}^{\widetilde{\lambda}_{it}},$$

where $\widetilde{\lambda}_{it}$ are cost-based Domar weights that measure the share of primary-factor costs flowing through firm $i$ to final demand. By construction, $\mu_t^\star$ captures the cumulative price–cost wedge faced by households after accounting for the full propagation of markups through the production network.

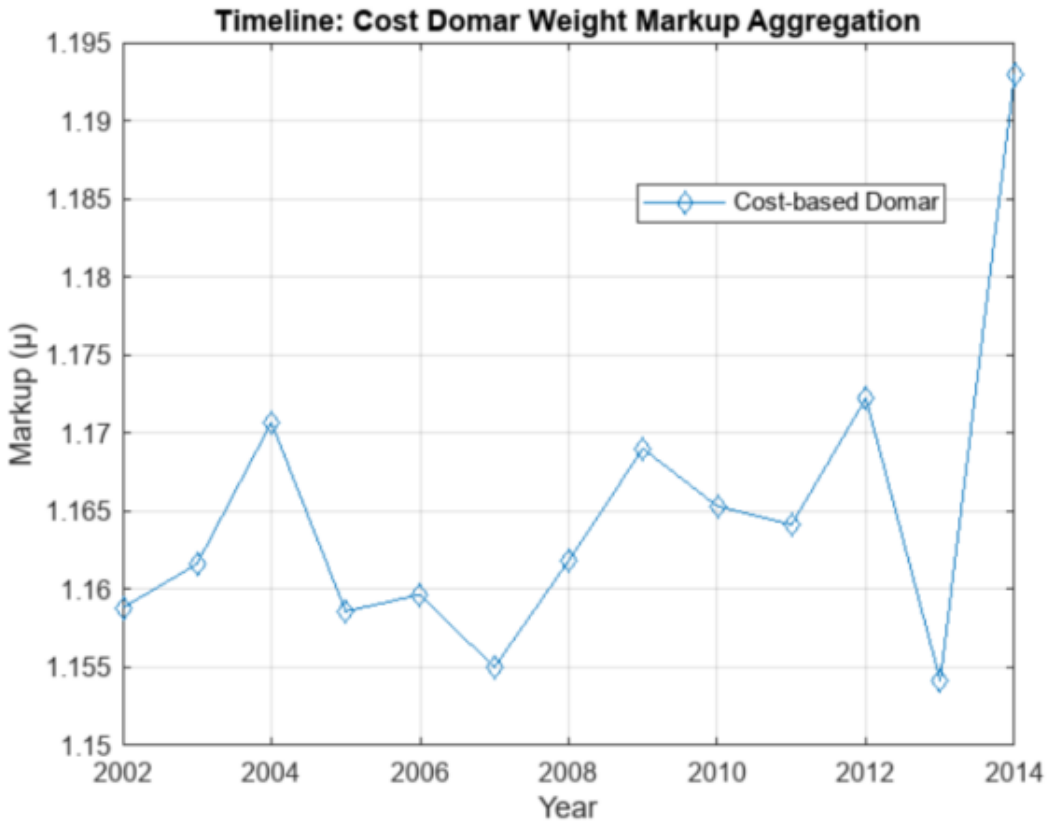



Figure 2: Cost-based Domar aggregate markup over time

Figure 2 reports the resulting aggregate markup for the Belgian economy over the period 2002–2014. The level is modest, fluctuating between approximately 1.16 and 1.19, and the time series is remarkably stable. There is a slight upward trend over the sample period, consistent with earlier evidence for Belgium (De Loecker, Fuss, and Van Biesebroeck 2018).

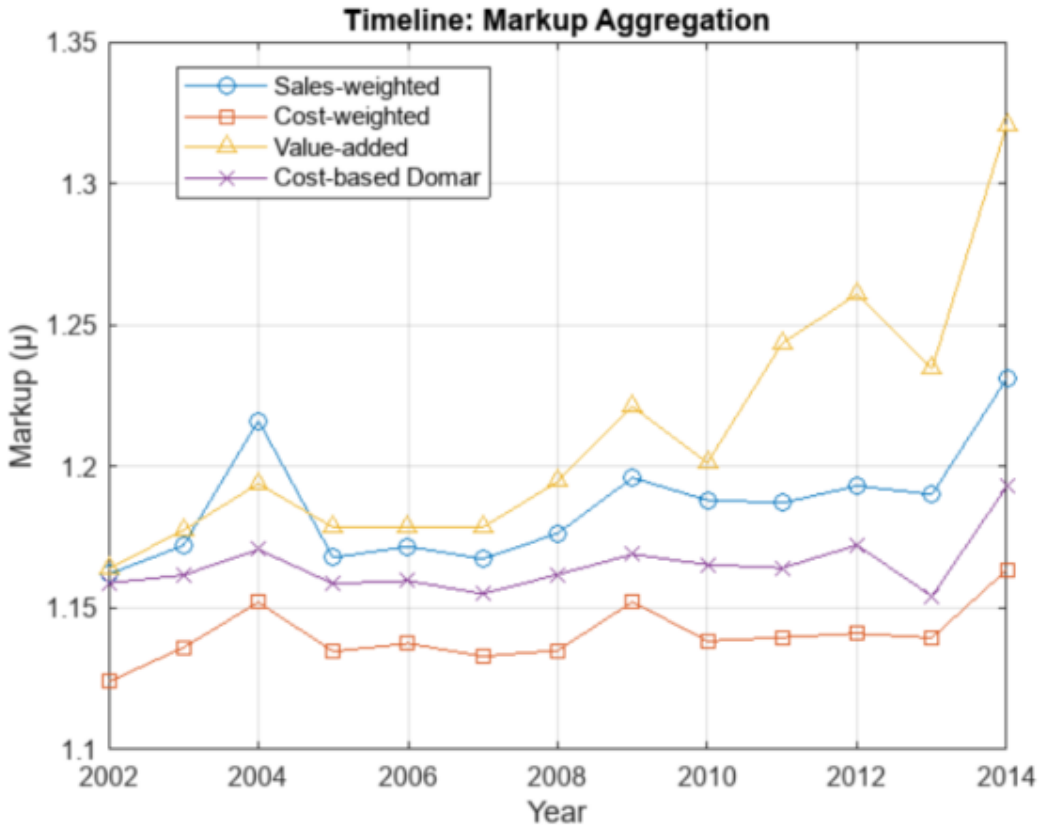



Figure 3: Markup aggregation across alternative weighting schemes

Figure 3 compares $\mu_t^\star$ to three standard benchmark aggregators: the sales-weighted, cost-weighted, and value-added-weighted averages. The purpose of this comparison is not to treat these measures as alternative estimates of the same underlying object, but to show how different weighting schemes map firm-level markups into distinct aggregate statistics.

Two results stand out.

First, all four series display very similar time-series dynamics. This indicates that aggregate fluctuations in markups are driven primarily by common movements in firm-level markups rather than by changes in the weighting scheme or in the structure of the production network.

Second, the levels differ systematically across aggregators. The cost-weighted average lies below $\mu_t^\star$, while the sales- and value-added-weighted averages are somewhat higher. Given that the aggregate markup itself is modest, these differences are economically meaningful. More importantly, the various aggregators rely on distinct weighting objects and therefore correspond to different economic concepts. By contrast, $\mu_t^\star$ is uniquely pinned down by household incidence and recovers the cumulative price–cost wedge implied by the production network.

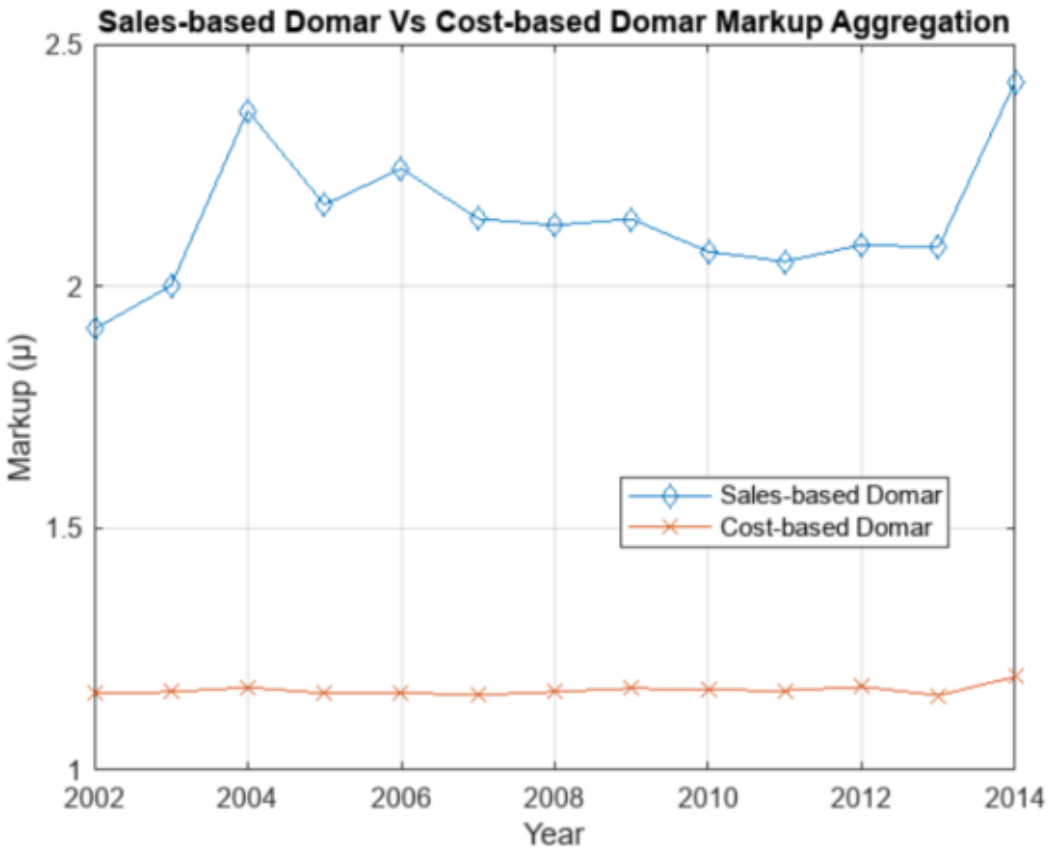



Figure 4: Cost-based Domar versus sales-based Domar aggregation

Figure 4 sharpens this point by comparing the cost-based Domar aggregator to its revenue-based counterpart. This is the most informative comparison for understanding why economic incidence matters. Both measures incorporate network structure, but they differ in the notion of exposure they use. The cost-based measure tracks the propagation of primary-factor costs to final demand, whereas the sales-based Domar measure tracks gross marked-up flows through the production network.

The resulting difference is very large. While $\mu_t^\star$ remains close to 1.17 throughout the sample, the sales-based Domar aggregator exceeds 2 in every year. This gap shows that incorporating network structure is not sufficient by itself: what matters is whether the weights correspond to household incidence or to gross-output exposure. Revenue-based Domar aggregation overstates the aggregate markup because it assigns too much importance to firms whose output passes repeatedly through downstream stages of production.

Overall, the evidence in this subsection delivers two conclusions. First, the network-consistent aggregate markup for Belgium is moderate in level and stable over time. Second, the economically relevant distinction is not simply between network and non-network aggregation, but between cost-based and revenue-based notions of exposure. Only the former yields an aggregate markup that

corresponds to the cumulative price–cost wedge faced by households.

### 5.3 Comparing Cost-Based Domar Aggregation to Sales-to-Final-Demand Aggregation

To isolate the role of upstream production layers, we compare the exact network aggregator $\mu_t^\star$ to a benchmark that only captures markups applied at the final stage of production. Specifically, we construct the geometric average of firm-level markups weighted by sales to final demand:

$$\mu_t^{FD,G} = \prod_{i \in FD} \mu_{it}^{b_{it}}.$$

This "sales-to-final-demand" aggregator captures only the direct markups embedded in goods purchased by households and excludes any markups embodied in intermediate inputs along the production chain.

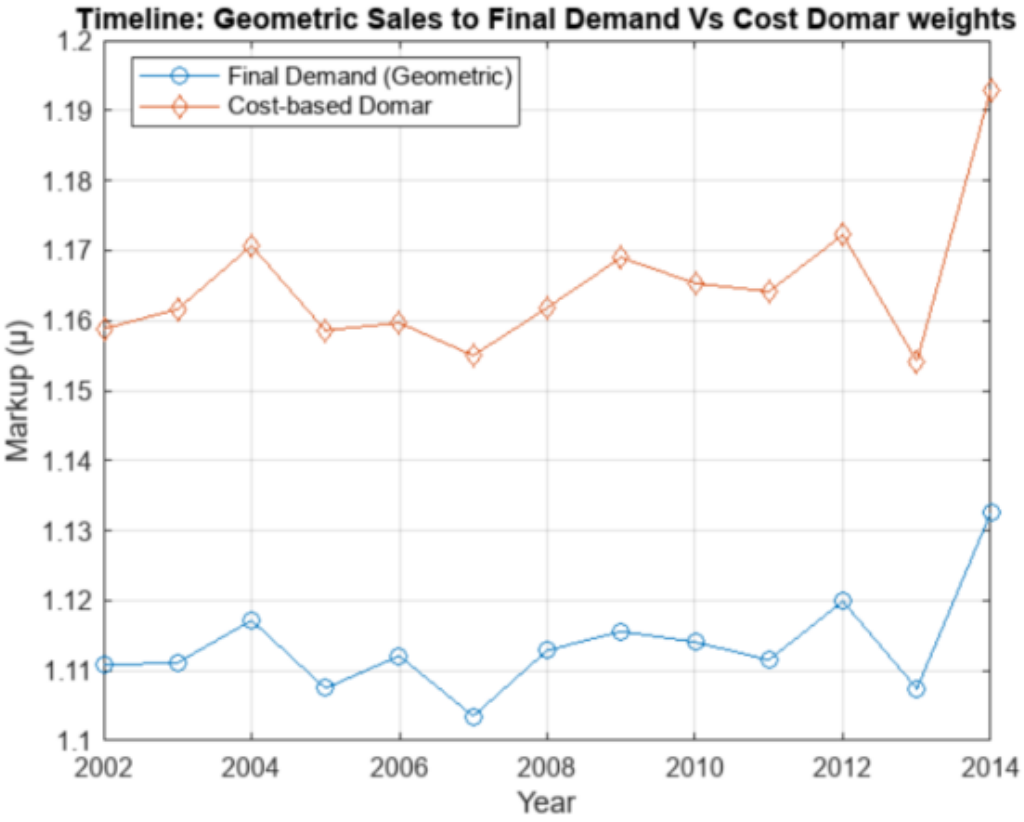



Figure 5: Cost-based Domar versus final-demand aggregation

Figure 5 shows that the two series display very similar time-series dynamics. This indicates that aggregate fluctuations in markups are primarily driven by firms located at or near the final stage of production.

At the same time, $\mu_t^\star$ is systematically higher than $\mu_t^{FD,G}$ by roughly 4 to 6 percentage points in each year. This gap reflects the contribution of upstream markups embodied in intermediate inputs and propagated through production chains.

This difference is economically meaningful. Relative to a total price–cost wedge of about 16 percentage points, a gap of 4 to 6 percentage points implies that approximately one quarter to one third of the aggregate markup originates upstream. Thus, while distortions are concentrated near final demand, upstream production layers account for a quantitatively significant share of the cumulative markup faced by households.

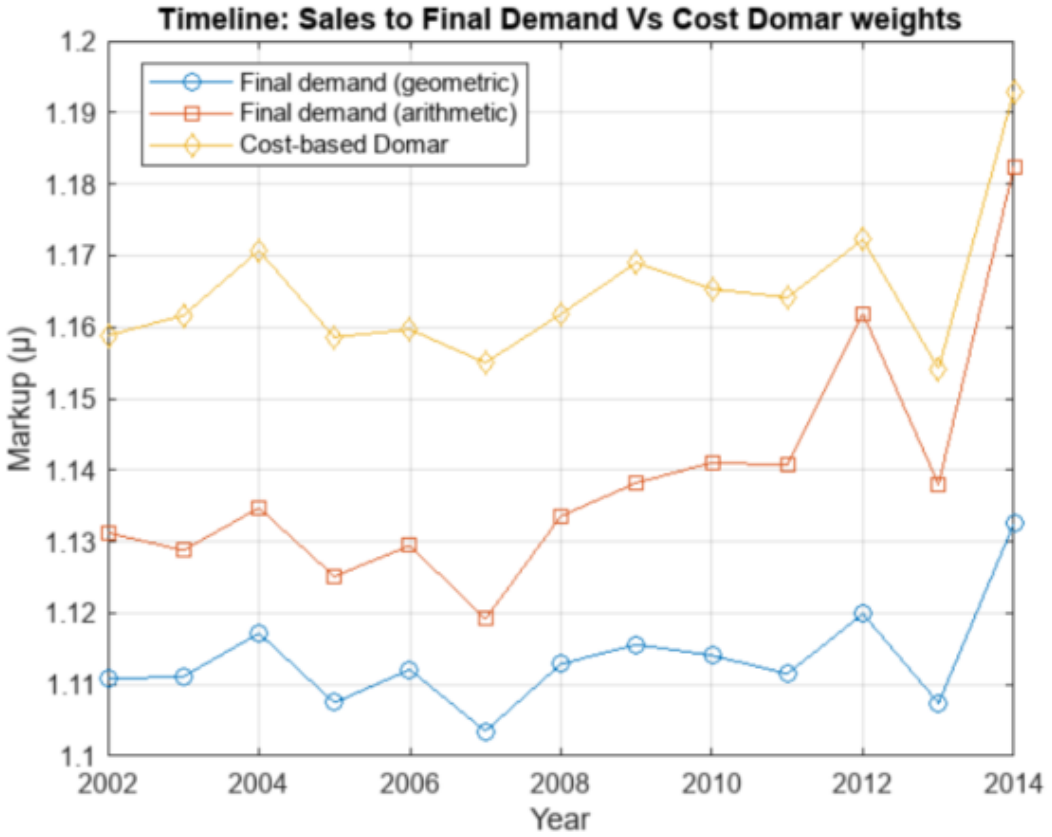



Figure 6: Cost-based Domar versus geometric and arithmetic final-demand aggregation

For comparison, Figure 6 also reports the arithmetic final-demand aggregator $\mu^{FD,A}$. Unlike the geometric measure, $\mu^{FD,A}$ does not respect the multiplicative nature of markup compounding and therefore does not map systematically into $\mu_t^\star$. This is consistent with the theoretical results in Section 3, which show that aggregate markups must be constructed using a geometric aggregation to account for the propagation of distortions along production chains.

### 5.4 Decomposing the Cost-Based Domar Markup Aggregator

We now decompose the aggregate markup into contributions from different distances to final demand. This decomposition makes it possible to quantify how much of the aggregate price–cost wedge originates at the point of final sale versus upstream production layers.

Starting from the aggregation formula

$$\ln \mu_t^\star = \mathbf{b}_t' \widetilde{\Psi}_t(\mu) \ln \mu_t,$$

and using the Neumann expansion $\widetilde{\Psi}_t = \sum_{L=0}^{\infty} \widetilde{\Omega}_t^L$, we obtain

$$\ln \mu_t^\star = \sum_{L=0}^{\infty} b_t' \widetilde{\Omega}_t^L \ln \mu_t,$$

where $L = 0$ captures firms selling directly to final demand, and $L \geq 1$ captures upstream contributions at increasing production distances.

To quantify the relative importance of each layer, we compute

$$\Upsilon_{t,L} \equiv \frac{b_t' \widetilde{\Omega}_t^L \ln \mu_t}{\ln \mu_t^\star}, \qquad \sum_{L \geq 0} \Upsilon_{t,L} = 1.$$

The coefficient $\Upsilon_{t,L}$ measures the share of the aggregate markup accounted for by firms located $L$ steps upstream from final demand. In practice, we truncate the decomposition at a finite horizon and group higher-order terms into a residual component. Because the spectral radius of $\widetilde{\Omega}_t$ is below one, the series converges rapidly and the residual is small.

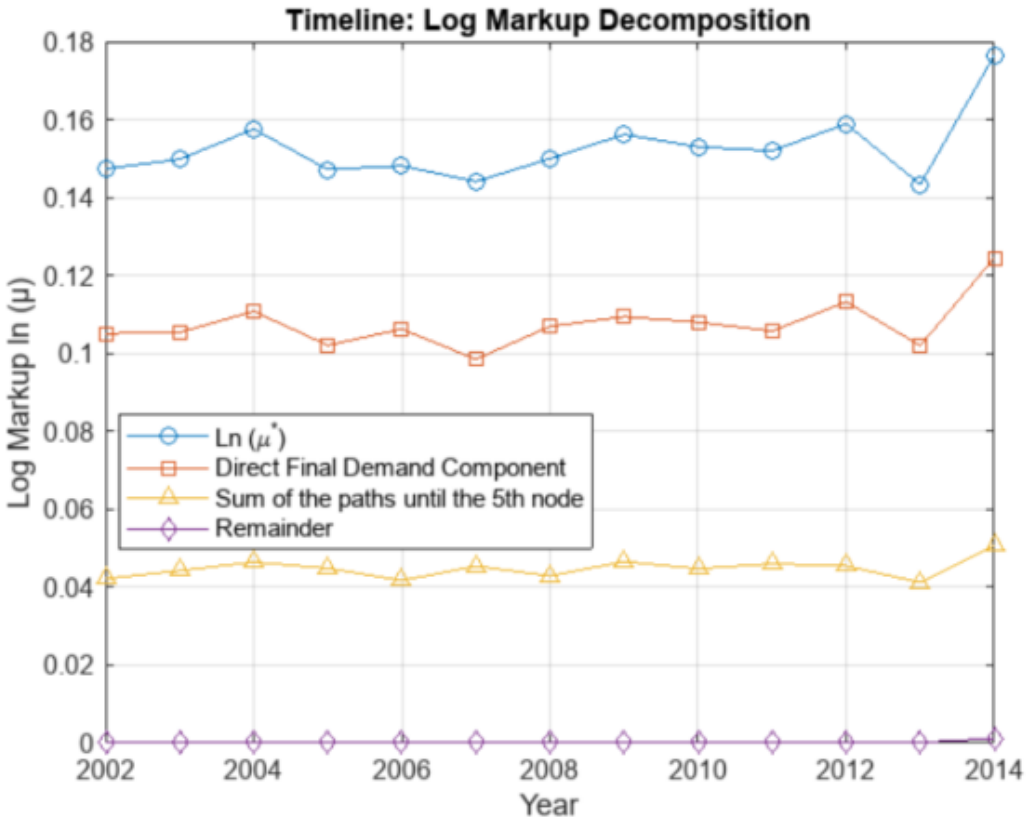



Figure 7: Contribution of different production layers to the aggregate markup. The direct component ($L = 0$) reflects firms selling to final demand, while upstream layers capture the propagation of markups through the production network.

Table 2: Decomposition of the aggregate markup by production depth

| Year | $L = 0$ | $L = 1$ | $L = 2$ | $L \geq 3$ |
|---|---|---|---|---|
| 2002 | 0.713 | 0.194 | 0.064 | 0.028 |
| 2003 | 0.703 | 0.193 | 0.069 | 0.033 |
| 2004 | 0.703 | 0.188 | 0.072 | 0.034 |
| 2005 | 0.694 | 0.196 | 0.074 | 0.035 |
| 2006 | 0.717 | 0.188 | 0.064 | 0.029 |
| 2007 | 0.683 | 0.203 | 0.075 | 0.037 |
| 2008 | 0.713 | 0.193 | 0.064 | 0.029 |
| 2009 | 0.701 | 0.198 | 0.069 | 0.031 |
| 2010 | 0.706 | 0.196 | 0.066 | 0.030 |
| 2011 | 0.696 | 0.198 | 0.070 | 0.035 |
| 2012 | 0.713 | 0.188 | 0.066 | 0.032 |
| 2013 | 0.712 | 0.193 | 0.065 | 0.030 |
| 2014 | 0.705 | 0.184 | 0.069 | 0.035 |

*Note:* Entries report the share of the aggregate markup attributable to firms at different distances from final demand. $L = 0$ denotes firms selling directly to final demand, while $L \geq 1$ captures upstream contributions.

Figure 7 and Table 2 report the decomposition.

Three main insights emerge.

First, the direct component ($L = 0$), corresponding to firms selling to final demand, accounts for the majority of the aggregate markup—around 70 percent in all years. Distortions applied at the point of final sale therefore dominate the aggregate price–cost wedge.

Second, upstream production layers contribute a substantial share of the aggregate markup, but this contribution is highly concentrated in the first few layers. The immediate upstream layer ($L = 1$) accounts for roughly 20 percent of the aggregate markup, while the second layer ($L = 2$) contributes an additional 5 to 7 percent. Together, the first two upstream layers account for roughly one quarter to one third of the total markup.

Third, the contribution of more distant production layers declines rapidly with depth. Beyond $L = 2$, each additional layer contributes only marginally, and layers beyond $L \geq 5$ account for a negligible share of the aggregate markup, as shown in Table 6 in Appendix C. This rapid decay implies that the relevant upstream distortions are tightly concentrated close to final demand.

Taken together, these results show that the gap between the network-consistent aggregate markup $\mu_t^\star$ and final-demand-based measures is driven almost entirely by the first few upstream layers. Measures that focus only on final demand therefore miss a quantitatively important component of market power, but the relevant propagation remains limited in depth.

This pattern is consistent with the "smile curve" view of global value chains, which emphasizes that value added and rents tend to be concentrated in stages of production close to final demand or involving differentiation and intangible inputs (Baldwin and Ito 2021). In that literature, fabrication stages are typically characterized by intense competition and low margins, while downstream activities such as branding and distribution capture higher rents.

Our decomposition provides a network-based counterpart to this idea. While upstream firms contribute to aggregate markups through input–output linkages, the bulk of the cumulative markup is generated at or near the final stage of production, with only a limited number of upstream layers contributing materially to the aggregate wedge.

This interpretation is also consistent with models of sequential production and incomplete contracting, in which surplus is allocated unevenly along the value chain (Antràs and Chor 2013; Antràs 2020).

Overall, the evidence shows that aggregate markups are moderate in level, shaped primarily by distortions near final demand, and augmented by a quantitatively meaningful but tightly bounded contribution from upstream production layers.

# 6 Conclusion

This paper develops a micro-founded framework for aggregating firm-level markups into an economy-wide measure of market power in production networks. The central challenge is that markups compound along supply chains: upstream firms raise downstream costs, and those higher costs are then marked up again before goods reach final consumers. As a result, aggregate markups cannot be defined independently of the production network. Standard aggregation methods fail to capture this structure. Final-demand measures omit upstream distortions, while revenue-based measures place excessive weight on gross intermediate flows.

Our main result is that the economically relevant aggregate markup is the cumulative price–cost wedge implied by equilibrium input–output linkages. We show that this object takes the form of an exact multiplicative aggregator with cost-based Domar weights, which map firm-level markups into household incidence. The resulting measure, $\mu^\star$, is therefore not a reduced-form average but the unique aggregate markup consistent with the structure of production networks. This framework provides both a network-consistent measure of aggregate markups and a decomposition of the aggregate wedge by production depth.

Using near-universe Belgian administrative data linking firm-to-firm transactions to production accounts, we implement this framework and quantify the aggregate markup implied by the domestic production network. Three empirical findings emerge.

First, aggregate markups in Belgium are modest and stable over time, fluctuating in a narrow range between approximately 1.16 and 1.19 over the period 2002–2014. The different aggregation

methods display very similar time-series dynamics, indicating that the main variation in aggregate markups is common across weighting schemes.

Second, the choice of aggregation method matters quantitatively because different measures correspond to fundamentally different economic objects. Standard sales-, cost-, and value-added-weighted averages yield aggregate markups that differ systematically from $\mu^\star$, while the geometric final-demand measure understates the aggregate markup by excluding upstream distortions. At the other extreme, revenue-based Domar aggregators substantially overstate aggregate markups because they place excessive weight on gross marked-up flows. The network-consistent aggregator isolates the economically relevant object by aligning weights with household incidence.

Third, decomposing the aggregate markup by production depth shows that most of the distortion arises at or near the final stage of production, while upstream markups still contribute a quantitatively meaningful share. Taken together, upstream production layers account for roughly one quarter to one third of the total price–cost wedge in level terms. Contributions from more distant layers are small, indicating that the propagation of distortions is largely concentrated close to final demand.

Taken together, these results imply that aggregate markups cannot be inferred from firm-level distributions alone. The production network determines how distortions propagate and accumulate, and therefore pins down the mapping from micro markups to the economy-wide price–cost wedge. Aggregate market power is thus inherently a network object: ignoring input–output linkages leads to systematic mismeasurement of both its level and its sources.

From a policy perspective, these findings imply that market power cannot be fully assessed by focusing solely on firms that sell directly to consumers. Because markups propagate through production networks, upstream firms contribute to the price–cost wedge faced by households even when they are not directly visible in final markets. Measures based only on final sales therefore understate the true extent of distortions, while approaches based on gross revenues overstate them. More broadly, the economic impact of a firm's markup depends not only on its level, but also on its position within the production network and the extent to which its output is used in downstream production.

Several avenues for future work are natural. One is to study how aggregate markups respond to shocks by combining the network-based aggregation framework with models of firm-level adjustment and reallocation. Another is to embed the aggregator into structural models of imperfect competition to quantify the welfare implications of upstream versus downstream market power. Finally, extending the analysis to cross-country data would help identify how differences in production-network structure shape aggregate markups across economies.

Overall, this paper shows that the relevant aggregate markup is the network-implied price–cost wedge faced by households, not a reduced-form average of firm-level markups. By providing a micro-founded and empirically implementable framework, it offers a unified way to measure and interpret market power in interconnected production systems. Understanding aggregate distortions therefore

requires not only measuring firm-level markups, but also tracing how they propagate through the structure of production.

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# Appendix

This appendix collects proofs of the theoretical results, additional derivations, and details on the empirical implementation.

# A Proofs

## A.1 Standing Assumptions

Throughout the appendix, we maintain the following conditions. First, the matrix $\widetilde{\Omega}(\mu) = \text{diag}(\mu)\Omega$ has spectral radius strictly below one, so that the inverse $(I - \widetilde{\Omega}(\mu))^{-1}$ exists. Second, all vectors and matrices are nonnegative and conformable, and final-demand weights $\mathbf{b}$ are strictly positive. These conditions ensure that the aggregation formulas and Neumann expansions used throughout are well defined.

## A.2 Useful Identities

For any markup vector $\mu$ such that $(I - \widetilde{\Omega}(\mu))$ is invertible, the cost-based Leontief inverse satisfies

$$d\widetilde{\Psi} = \widetilde{\Psi}\,(d\widetilde{\Omega})\,\widetilde{\Psi}, \tag{1}$$

$$\widetilde{\Psi}\,\widetilde{\Omega} = \widetilde{\Psi} - I. \tag{2}$$

The first identity follows from differentiating the matrix inverse, and the second follows directly from $\widetilde{\Psi} = (I - \widetilde{\Omega})^{-1}$.

## A.3 Proof of Proposition 1

**Intuition.** The result follows from differentiating the aggregate markup with respect to firm-level markups and showing that general equilibrium feedback effects cancel exactly, leaving only direct exposure weights given by cost-based Domar weights.

Let

$$\widetilde{\Omega}(\mu) \equiv \text{diag}(\mu)\,\Omega, \qquad \widetilde{\Psi}(\mu) \equiv (I - \widetilde{\Omega}(\mu))^{-1}.$$

Define

$$\mathbf{z} \equiv \widetilde{\Psi}(\mu)\,\mathbf{w}, \qquad \mathbf{p} \equiv \text{diag}(\mu)\,\mathbf{z}, \qquad F \equiv \mathbf{b}'\mathbf{z}, \qquad E \equiv \mathbf{b}'\mathbf{p}.$$

Then $\mu^\star = E/F$, and the cost-based Domar weights satisfy

$$\widetilde{\boldsymbol{\lambda}}' = \mathbf{b}'\widetilde{\Psi}(\mu).$$

We consider a small proportional change $d\ln\mu$, so that

$$d\mu = \text{diag}(\mu)\,d\ln\mu.$$

Differentiating $E = \mathbf{b}'\text{diag}(\mu)\mathbf{z}$ and $F = \mathbf{b}'\mathbf{z}$ yields

$$dE = \mathbf{b}'\text{diag}(\mu)\, d\mathbf{z} + \mathbf{b}'\text{diag}(d\mu)\, \mathbf{z}, \qquad dF = \mathbf{b}'\, d\mathbf{z}.$$

Dividing by levels gives

$$d\ln E = \frac{\mathbf{b}'\text{diag}(\mu)\, d\mathbf{z} + \mathbf{b}'\text{diag}(\mu)\text{diag}(d\ln\mu)\, \mathbf{z}}{E}, \qquad d\ln F = \frac{\mathbf{b}'\, d\mathbf{z}}{F},$$

where

$$d\mathbf{z} = d\widetilde{\Psi}\,\mathbf{w} = \widetilde{\Psi}(d\widetilde{\Omega})\widetilde{\Psi}\mathbf{w} = \widetilde{\Psi}(d\widetilde{\Omega})\mathbf{z}.$$

Subtracting gives

$$d\ln\mu^\star = \frac{\mathbf{b}'\text{diag}(\mu)\widetilde{\Psi}(d\widetilde{\Omega})\mathbf{z}}{E} - \frac{\mathbf{b}'\widetilde{\Psi}(d\widetilde{\Omega})\mathbf{z}}{F} + \frac{\mathbf{b}'\text{diag}(\mu)\text{diag}(d\ln\mu)\mathbf{z}}{E}.$$

Now substitute

$$d\widetilde{\Omega} = \text{diag}(\mu)\text{diag}(d\ln\mu)\Omega,$$

and use

$$\widetilde{\Psi}\widetilde{\Omega} = \widetilde{\Psi} - I.$$

After substituting and using $\widetilde{\Psi}\widetilde{\Omega} = \widetilde{\Psi} - I$, the two terms involving $d\widetilde{\Omega}$ cancel exactly. This cancellation reflects the fact that general equilibrium feedback effects through input–output linkages offset each other, leaving only the direct exposure term. Then,

$$d\ln\mu^\star = \frac{\mathbf{b}'\text{diag}(\mu)\text{diag}(d\ln\mu)\mathbf{z}}{E}.$$

Using $\mathbf{p} = \text{diag}(\mu)\mathbf{z}$ and $E = \mathbf{b}'\mathbf{p}$, this becomes

$$d\ln\mu^\star = \mathbf{b}'\widetilde{\Psi}(\mu)\, d\ln\mu = \sum_i \widetilde{\lambda}_i\, d\ln\mu_i.$$

To obtain the global expression, integrate along the homothetic path

$$\mu(t) = \exp(t\ln\mu), \qquad t \in [0,1],$$

which connects $\mu = \mathbf{1}$ to $\mu$. Since $\mu^\star(\mathbf{1}) = 1$, integrating yields

$$\ln\mu^\star = \sum_i \widetilde{\lambda}_i \ln\mu_i, \qquad \mu^\star = \prod_i \mu_i^{\widetilde{\lambda}_i}.$$

□

### A.4 Proof of Proposition 2

**Intuition.** The result follows from expanding the markup-adjusted Leontief inverse into a Neumann series, which maps production chains into contributions by path length.

From Proposition 1, the aggregate markup satisfies

$$\ln \mu^\star = \mathbf{b}' \widetilde{\Psi}(\mu) \ln \mu, \qquad \widetilde{\Psi}(\mu) = (I - \widetilde{\Omega}(\mu))^{-1},$$

where

$$\widetilde{\Omega}(\mu) \equiv \operatorname{diag}(\mu)\, \Omega.$$

We maintain that $\widetilde{\Omega}(\mu)$ has spectral radius strictly below one, so that the inverse exists. Under this condition, the Neumann expansion applies:

$$\widetilde{\Psi}(\mu) = \sum_{L=0}^{\infty} \widetilde{\Omega}(\mu)^L.$$

Substituting yields

$$\ln \mu^\star = \sum_{L=0}^{\infty} \mathbf{b}' \widetilde{\Omega}(\mu)^L \ln \mu.$$

Define

$$\Upsilon_L = \frac{\mathbf{b}' \widetilde{\Omega}(\mu)^L \ln \mu}{\ln \mu^\star}.$$

Then $\sum_{L \geq 0} \Upsilon_L = 1$.

□

## B Derivations and Implementation

### B.1 From Aggregation to Path Decomposition

Proposition 1 expresses the aggregate markup as a cost-based Domar-weighted average of firm-level markups,

$$\ln \mu^\star = \mathbf{b}' \widetilde{\Psi}(\mu) \ln \mu.$$

Proposition 2 follows from applying the Neumann expansion to the cost-based Leontief inverse,

$$\widetilde{\Psi}(\mu) = \sum_{L=0}^{\infty} \widetilde{\Omega}^L, \qquad \widetilde{\Omega} = \operatorname{diag}(\mu)\, \Omega,$$

which yields a decomposition of the aggregate markup into contributions associated with production paths of different lengths.

## B.2 Interpretation of Higher-Order Terms

The matrix power $\widetilde{\Omega}^L$ captures the propagation of markups along production chains of length $L$. Each term represents the cumulative effect of markups applied by firms located $L$ steps upstream from final demand.

Because $\widetilde{\Omega}$ is a nonnegative matrix with spectral radius strictly below one under Assumption 1, the series converges and higher-order terms decay geometrically. This implies that distant production layers contribute progressively less to the aggregate markup.

## B.3 Empirical Implementation

The implementation takes firm-level markups $\mu_{it}$ as given and combines them with the observed input–output structure to construct $\mu_t^\star$. The aggregation itself is exact conditional on the network representation, while the validity of the resulting aggregate markup inherits the identifying assumptions underlying the estimation of firm-level markups.

In the empirical analysis, we compute the decomposition

$$\Upsilon_{t,L} = \frac{\mathbf{b}_t' \widetilde{\Omega}_t^L \ln \mu_t}{\ln \mu_t^\star}, \qquad L \geq 0,$$

which measures the share of the aggregate markup attributable to firms located $L$ steps upstream from final demand.

In practice, we truncate the infinite series at a finite horizon $L_{\max}$ and group the remaining higher-order terms into a residual component. Because the series converges rapidly, the residual is quantitatively small.

In the baseline results, we report the decomposition up to a finite number of layers and collect all remaining higher-order contributions in a residual term. Empirically, this residual is negligible, confirming that markup propagation is concentrated in the first few upstream stages of production.

# C Additional Tables

## Descriptive Statistics

Table 3: Aggregate Markups Under Alternative Weighting Schemes

| Year | Sales-Weighted | Cost-Weighted | Value-Added Weighted | Sales-Based Domar | Cost-Based Domar |
|---|---|---|---|---|---|
| 2002 | 1.162 | 1.124 | 1.1638 | 1.9115 | 1.1588 |
| 2003 | 1.172 | 1.136 | 1.1775 | 2.0009 | 1.1616 |
| 2004 | 1.2162 | 1.152 | 1.1939 | 2.3627 | 1.1707 |
| 2005 | 1.1678 | 1.1347 | 1.1785 | 2.1681 | 1.1586 |
| 2006 | 1.1717 | 1.1375 | 1.1785 | 2.2430 | 1.1597 |
| 2007 | 1.1673 | 1.1330 | 1.1785 | 2.1392 | 1.1550 |
| 2008 | 1.1764 | 1.1350 | 1.1946 | 2.1258 | 1.1618 |
| 2009 | 1.1960 | 1.1522 | 1.2212 | 2.1389 | 1.1690 |
| 2010 | 1.1881 | 1.1384 | 1.2013 | 2.0714 | 1.1653 |
| 2011 | 1.1872 | 1.1396 | 1.2436 | 2.0509 | 1.1641 |
| 2012 | 1.1932 | 1.1410 | 1.2608 | 2.0853 | 1.1722 |
| 2013 | 1.1901 | 1.1394 | 1.2346 | 2.0803 | 1.1542 |
| 2014 | 1.2310 | 1.1635 | 1.3206 | 2.4223 | 1.1930 |

*Notes:* The table reports aggregate markups under alternative weighting schemes. Sales-, cost-, and value-added-weighted measures correspond to standard firm-level aggregation. The sales-based Domar measure weights firms by total sales (double-counting intermediates), while the cost-based Domar measure corresponds to the exact network aggregator and accounts for the propagation of markups through production chains.

## Aggregation Weights

Table 4: Aggregate Markup: Cost-Based Domar Measure

| Year | Cost-Based Domar |
|---|---|
| 2002 | 1.1588 |
| 2003 | 1.1616 |
| 2004 | 1.1707 |
| 2005 | 1.1586 |
| 2006 | 1.1597 |
| 2007 | 1.1550 |
| 2008 | 1.1618 |
| 2009 | 1.1690 |
| 2010 | 1.1653 |
| 2011 | 1.1641 |
| 2012 | 1.1722 |
| 2013 | 1.1542 |
| 2014 | 1.1930 |

*Notes:* The table reports the aggregate markup computed using the cost-based Domar weights, which account for the full propagation of markups through the production network.

## Aggregation Comparisons

Table 5: Aggregate Markups: Final-Demand vs. Network-Based Measures

| Year | Final Demand (Geometric) | Final Demand (Arithmetic) | Cost-Based Domar |
|---|---|---|---|
| 2002 | 1.1108 | 1.1312 | 1.1588 |
| 2003 | 1.1111 | 1.1288 | 1.1616 |
| 2004 | 1.1172 | 1.1347 | 1.1707 |
| 2005 | 1.1075 | 1.1251 | 1.1586 |
| 2006 | 1.1121 | 1.1294 | 1.1597 |
| 2007 | 1.1035 | 1.1192 | 1.1550 |
| 2008 | 1.1129 | 1.1336 | 1.1618 |
| 2009 | 1.1156 | 1.1382 | 1.1690 |
| 2010 | 1.1141 | 1.1410 | 1.1653 |
| 2011 | 1.1115 | 1.1407 | 1.1641 |
| 2012 | 1.1199 | 1.1618 | 1.1722 |
| 2013 | 1.1074 | 1.1380 | 1.1542 |
| 2014 | 1.1326 | 1.1824 | 1.1930 |

*Notes:* The table reports alternative aggregate markup measures. The final-demand measures use only firms selling to households, while the cost-based Domar measure incorporates the full production network and captures upstream propagation of markups.

## Path Decomposition

Table 6: Path-Length Decomposition of the Aggregate Markup

| Year | $\ln \mu^\star$ | $L=0$ | $L=1$ | $L=2$ | $L=3$ | $L=4$ | $L=5$ | $L \geq 6$ |
|---|---|---|---|---|---|---|---|---|
| 2002 | 0.1474 | 0.1051 | 0.0286 | 0.0094 | 0.0030 | 0.0009 | 0.0003 | 0.0001 |
| 2003 | 0.1498 | 0.1054 | 0.0289 | 0.0104 | 0.0035 | 0.0011 | 0.0004 | 0.0002 |
| 2004 | 0.1576 | 0.1108 | 0.0297 | 0.0114 | 0.0036 | 0.0014 | 0.0004 | 0.0003 |
| 2005 | 0.1472 | 0.1021 | 0.0288 | 0.0109 | 0.0034 | 0.0013 | 0.0004 | 0.0003 |
| 2006 | 0.1481 | 0.1063 | 0.0279 | 0.0094 | 0.0031 | 0.0010 | 0.0003 | 0.0002 |
| 2007 | 0.1441 | 0.0985 | 0.0293 | 0.0108 | 0.0037 | 0.0012 | 0.0004 | 0.0002 |
| 2008 | 0.1500 | 0.1070 | 0.0289 | 0.0097 | 0.0031 | 0.0010 | 0.0003 | 0.0001 |
| 2009 | 0.1562 | 0.1094 | 0.0310 | 0.0107 | 0.0034 | 0.0011 | 0.0004 | 0.0002 |
| 2010 | 0.1530 | 0.1080 | 0.0299 | 0.0101 | 0.0033 | 0.0010 | 0.0004 | 0.0002 |
| 2011 | 0.1520 | 0.1057 | 0.0301 | 0.0106 | 0.0036 | 0.0012 | 0.0004 | 0.0002 |
| 2012 | 0.1589 | 0.1132 | 0.0299 | 0.0105 | 0.0034 | 0.0012 | 0.0004 | 0.0002 |
| 2013 | 0.1434 | 0.1020 | 0.0276 | 0.0093 | 0.0030 | 0.0009 | 0.0003 | 0.0002 |
| 2014 | 0.1764 | 0.1245 | 0.0325 | 0.0121 | 0.0041 | 0.0014 | 0.0006 | 0.0012 |

*Note:* The table reports the contribution of each production layer $L$ to the aggregate markup $\ln \mu^\star$. The component $L = 0$ corresponds to firms selling directly to final demand, while $L \geq 1$ captures upstream contributions at increasing distances from final demand. The remainder reflects contributions from paths of length $L \geq 6$.